

Lab 1 – DNA Barcoding of Sashimi

Designing PCR primers, identifying a mystery fish, and estimating genetic diversity in R

Course	PUBH 6860 – Principles of Bioinformatics (Fall 2026)
Assignment	Lab 1 – DNA Barcoding of Sashimi
Aligned session	Molecular Biology & PCR; Sequence Databases
Required reading	Chapters 1 (Molecular Biology) & 6 (Sequence Databases)
Points	100 points (Labs = 35% of course grade)
Assigned / Due	Distributed in class • Due the following week by 11:59 pm
Submission	Upload a single PDF to Blackboard. Late work: –5%/day up to 5 days.
Collaboration	Group work on the analysis is encouraged; each student submits their own write-up.

Overview & Purpose

DNA barcoding uses a short, standardized gene region to tell species apart. In animals, the go-to barcode is a portion of the mitochondrial cytochrome c oxidase subunit I (COI) gene, which varies enough between species to distinguish them while staying consistent within a species. Seafood is a classic application: menu names do not always match what is on the plate, and a barcode can settle the question.

In this lab you will play the inquisitive biologist at a sushi bar. You will design PCR primers to amplify a query sequence, compare it against the candidate menu species, estimate pairwise genetic diversity in R, and decide what you actually ate — then report your work as a short scientific paper. As in all lab write-ups, **document your tools and parameters so another person could reproduce your analysis.**

Learning Objectives

By completing this lab, you will be able to:

1. Explain the principle of DNA barcoding and the role of the mitochondrial COI gene in species identification. (*Chapter 1*)
2. Design PCR primers that specifically amplify a target DNA region. (*Chapter 1*)
3. Retrieve reference sequences from GenBank and compare a query against candidates using BLAST. (*Chapter 6*)
4. Compute and interpret percent sequence identity to assign a sample to a species. (*Chapter 6*)
5. Estimate pairwise genetic diversity in R and interpret it, accounting for differences in sequence length. (*Course objective: apply analytical software to biomedical data*)
6. Communicate methods and results as a scientific paper (journal *Bioinformatics* style), documenting tools for reproducibility. (*Course objective: communicate scientific results*)

The Scenario

You and some friends have just had dinner at a local sushi bar. Being the inquisitive biologist, you wonder if your group really ate what was listed on the menu. So you save a bit of tissue from your sashimi. The menu lists the following possibilities: “Hirame” (flounder), “Sake” (salmon), “Hamachi” (yellowtail), and “Maguro” (tuna). **You ordered the Maguro.** Using the DNA sequence recovered from your sample (below), design PCR

primers to amplify the exact sequence, compare it against the candidate species, and determine what you actually ate.

Your query sequence (5'→3'):

```
CGGGCAGAACTAAGCCAGCCCGGCTCTCTTCTCGGAGACGACCAATCTATAATGTAATTGTTACAGCAC
ATGCTTTTCGTAATAATTTCTTTATAGTAATACCAATTATGATTGGAGGCTTTGGAACTGACTAGTACC
CCTCATGATTGGTGACACAGACATGGCCTTCCCTCGAATAAATAACATGAGCTTTTGACTTCTCCCCCCC
TCATTTCTTCTTCTCTCGCCTCATCTGGAGTTGAAGCAGGTGCCGGCACAGGATGGACTGTTTATCCCC
CGCTCGCAGGCAATCTTGCCACGCTGGACCTTTTGTGACTTAACCATATTCTCCCTCCACTTGGCCGG
AGCGTCATCTATTTAGGTGCAATTAATTTATCACCAACATTATTAACATGAAACCCCTGCCATCTCC
CAATATCAAACACCCATATTTGTGTGATCCGTCCTAATTACCGCAGTACTACTCCTTCTAT
```

Length: 481 nucleotides. Treat the sequence as case-insensitive.

Your Tasks

1. Design PCR primers that would amplify the exact sequence above; report the forward and reverse primers (5'→3').
2. Retrieve candidate reference sequences for the menu species — Hiramé (flounder), Sake (salmon), Hamachi (yellowtail), and Maguro (tuna) — from GenBank, and compare your query to each (e.g., relative percent identity).
3. Determine the species your sample most likely came from, and state whether it matches what you ordered (Maguro).
4. Estimate pairwise genetic diversity in R across the sequences you gathered (your query plus the candidate sequences); report and interpret the results — see the guidance below.
5. Write up your approach and findings as a short scientific paper following the Write-Up Format below.

Estimating Pairwise Genetic Diversity in R

Beyond a single percent-identity value, quantify the genetic diversity among the sequences you gathered. Assemble your query and the candidate reference sequences into one **FASTA file**, and use R to estimate pairwise genetic diversity (pairwise distances and, optionally, nucleotide diversity).

What to submit

- A pairwise distance matrix (and/or nucleotide diversity, π) computed in R from your FASTA file, plus the R code you ran.
- A short interpretation linking the diversity values to your species identification.

Example — adapt the file name to your own data:

```
library(ape)
# Put your query + candidate species sequences into one FASTA, then import:
seqs <- read.dna("sashimi_sequences.fasta", format = "fasta")

D <- dist.dna(seqs, model = "raw", pairwise.deletion = TRUE) # per-site p-distances
D                                     # pairwise diversity = proportion of sites that differ

library(pegas)
nuc.div(seqs)                # nucleotide diversity (pi): mean pairwise differences per site
```

Questions to address

- What are the pairwise genetic distances (diversity) among your sequences? Which candidate is least diverged from your query, and is that consistent with your species call?
- Did your sequences differ in length? Is your diversity estimate a **per-site proportion** (a p-distance or π) or a raw count of differences?
- Explain how comparing sequences of different lengths using a raw count of differences could distort your results and your identification. **How did you adjust for different sequence lengths?**

Write-Up Format

Lab write-ups follow the standard form of a scientific paper, specifically the format of the journal *Bioinformatics* (<https://academic.oup.com/bioinformatics>), the leading journal in the field. Begin with an original title, followed by your name, the course, and the date. Papers should be typed, single-spaced, and in 12 pt font, and include the following sections.

Required sections

- **Title** — a concise but informative summary of the report.
- **Introduction** — state the general problem and provide background context, preparing the reader and convincing them the problem is worth addressing.
- **Materials and Methods** — describe how you designed primers, obtained reference data, compared sequences, and ran the diversity analysis in R. Explicitly name and cite bioinformatic tools and give the parameters used.
- **Results** — present your primers, the percent-identity comparison, and the pairwise diversity/distance matrix (a table or figure is ideal). Present results only; save interpretation for the Discussion.
- **Discussion** — interpret your results; draw conclusions about the species identity and any mislabeling; address whether your diversity estimate was adjusted for sequence length; synthesize your ideas and note limitations.
- **Conclusion** — wrap up and summarize. This may be folded into the end of the Discussion rather than a separate section.
- **References (Literature Cited)** — list the literature and tools used, in the format of the journal *Bioinformatics*. Failure to cite work used may be considered plagiarism.

Suggested Tools & Resources

You may use any reputable software, but the tools below are recommended. Whatever you use, record the tool name, version, and parameters so your work can be reproduced.

- **Primer design:** Primer3 (<https://primer3.ut.ee/>) or NCBI Primer-BLAST.
- **Database & search:** GenBank / NCBI Nucleotide (<https://www.ncbi.nlm.nih.gov/nucleotide/>) and blastn (<https://blast.ncbi.nlm.nih.gov/>).
- **Genetic diversity in R:** the *ape* package (*read.dna*, *dist.dna*) and *pegas* (*nuc.div*); *seqinr* for reading/handling FASTA.
- **FASTA handling:** a plain-text editor (VS Code, Sublime, Notepad++, BBEdit).

Frequently Asked Questions

Do I need to design primers that work for all possible fish species?

No. For this basic exercise, assume you only need to design primers that would amplify the exact sequence provided above.

How detailed do I need to be about the DNA extraction and PCR process?

Only minimally. Provide your primer sequences, but — because this is a bioinformatics course — focus on the bioinformatic details. For the extraction/PCR you can simply write something like, “We extracted DNA from the sample collected from... and conducted a PCR reaction with the following primers...”

How should I handle sequences of different lengths when estimating diversity?

Use a per-site measure — the proportion of sites that differ (a p-distance) or nucleotide diversity (π) — rather than a raw count of differences, so that sequences of different lengths are compared on a common, per-site basis. State clearly which measure you used and why.

Submission & Formatting Expectations

- Write the report as a scientific paper with the sections above, in order, with an original title, your name, the course, and the date.
- Report your forward and reverse primers, and state the tools, versions, and parameters used so the analysis can be reproduced.
- Present the percent-identity comparison and the pairwise diversity matrix as labeled tables or figures, and include your R code.
- Acknowledge any use of generative AI — the tool and version, the prompts supplied, and what was AI-generated versus your own contribution (see the GW University Policy on GAI).
- Save the write-up as a single PDF and upload it to Blackboard by 11:59 pm on the due date. Typed, single-spaced, 12 pt font; label any tables or figures.